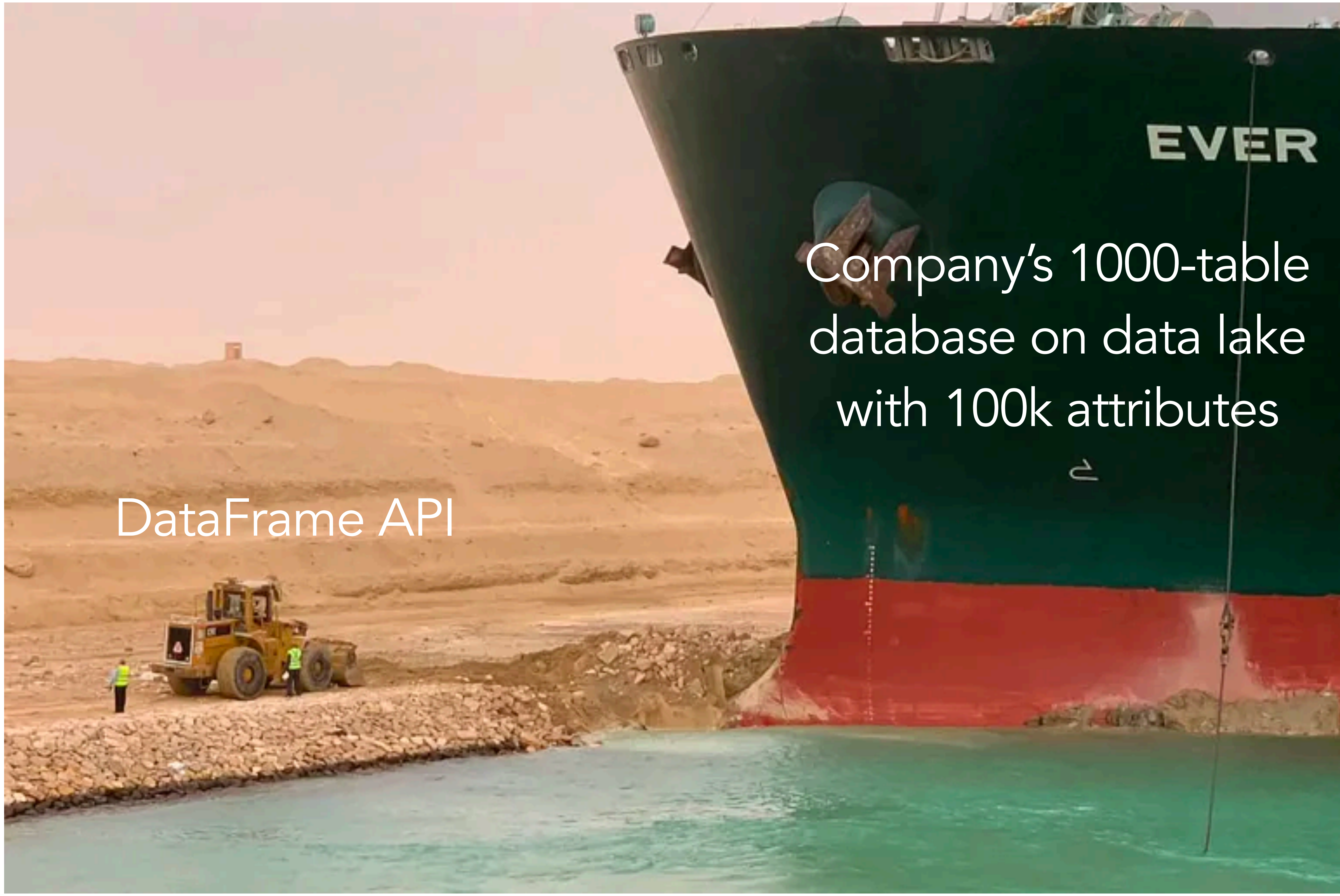
DSC 204a Scalable Data Systems

- Haojian Jin



Meme idea credit: https://datasystemsfun.tumblr.com/

Published Preview Assign To Edit

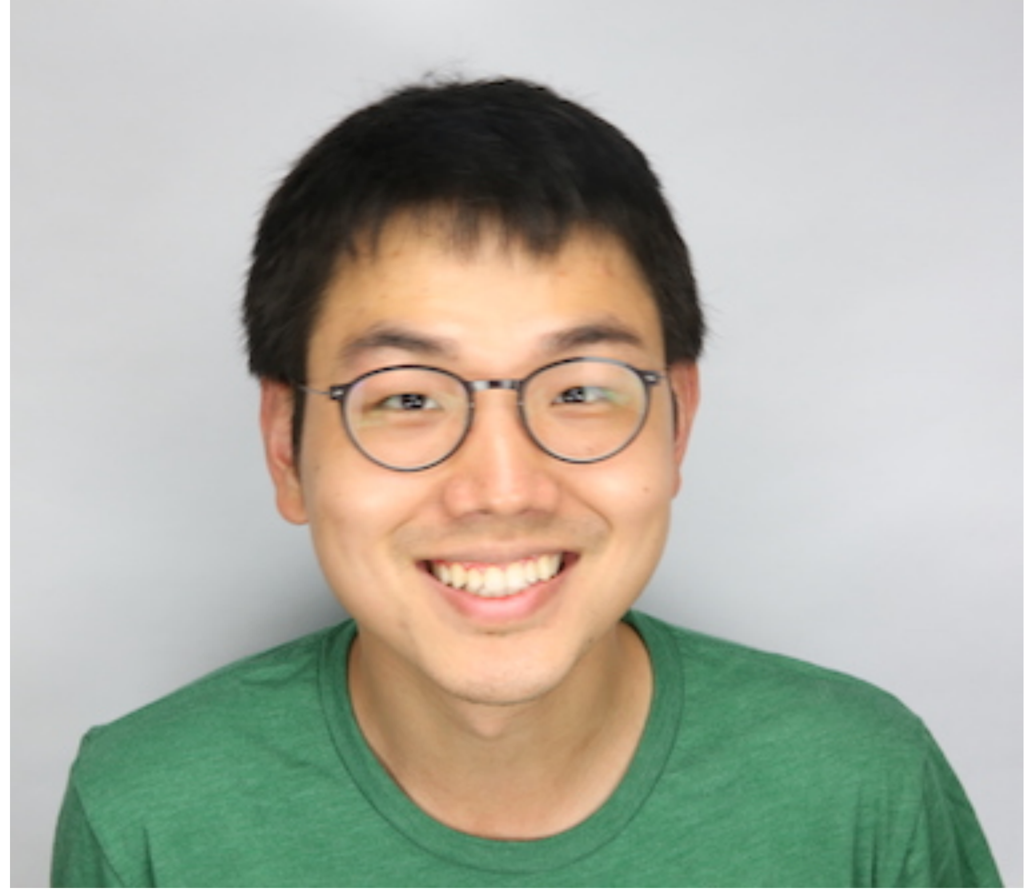
Pre-course quizz

Students who have taken the quiz (17)	Students who haven't taken the quiz (34)

[redacted]

This is a short multiple-choice quiz to help me understand your background so I can adjust the course content for this quarter. There are no right or wrong answers for grading purposes -- just answer to the best of your ability.

Bio



Haojian Jin (http://haojianj.in/)

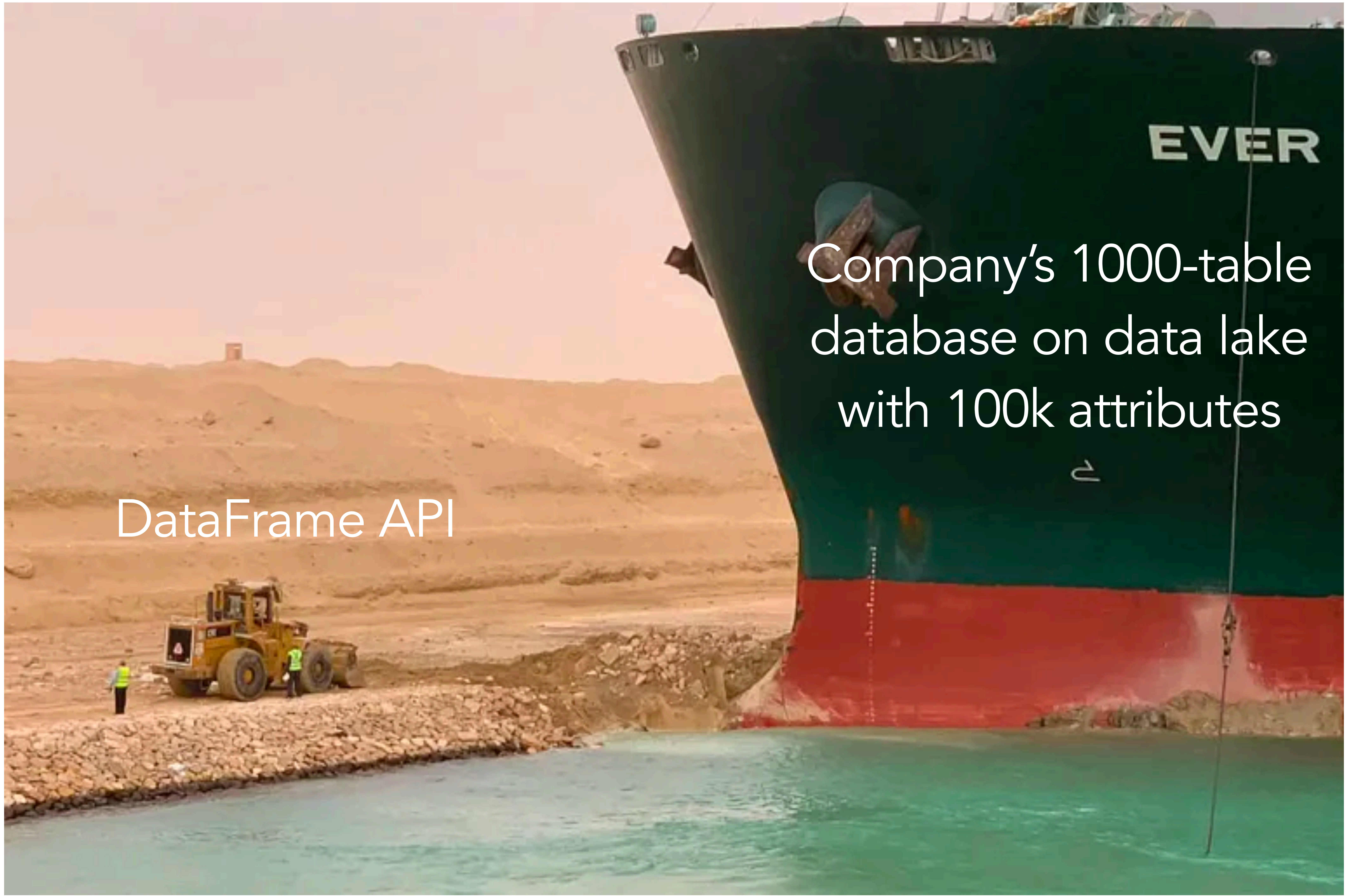
Asst. Prof @ UCSD-HDSI

AI Smith Lab

Ph.D. from CMU Human-Computer Interaction Institute

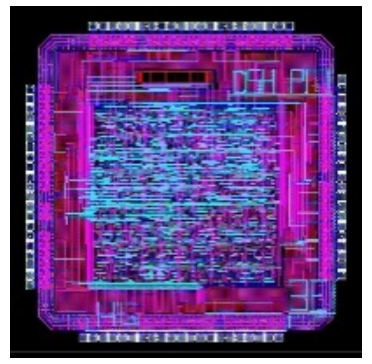
Before Ph.D.: worked at Yahoo Research, ran a startup

What is this course about?

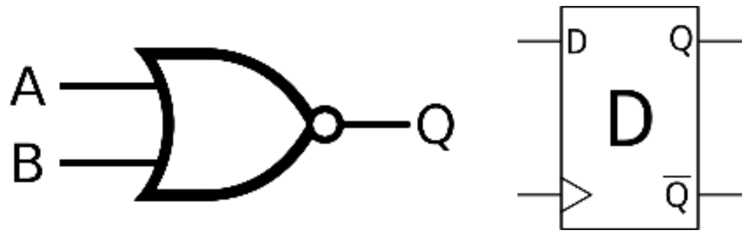


Levels of Abstraction

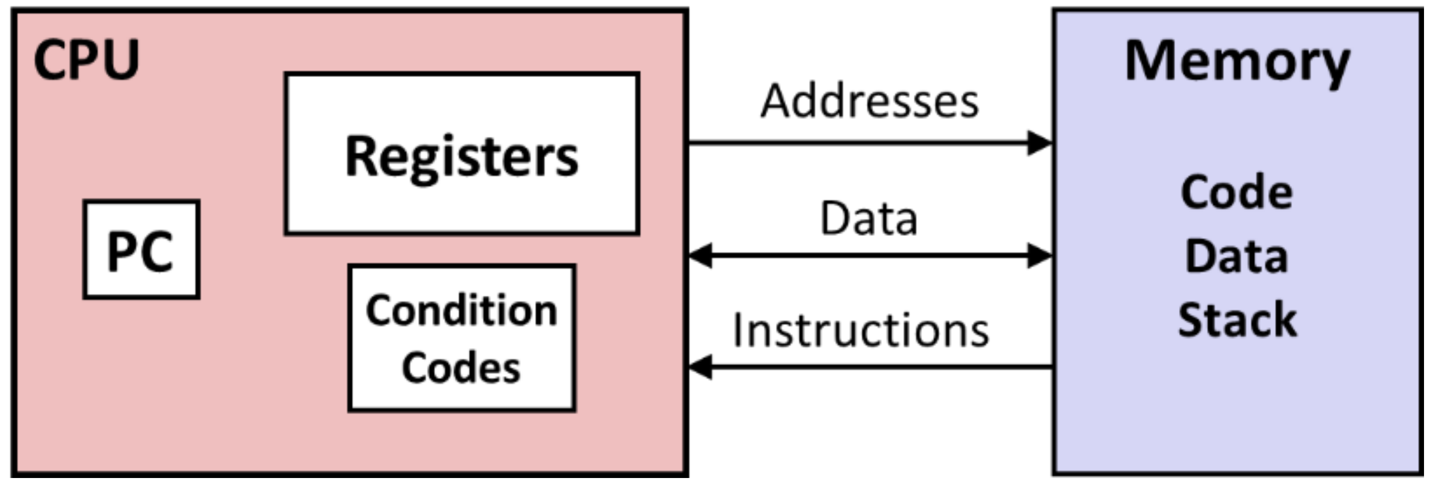
Computer Designer



Gates, clocks, circuit layout, …



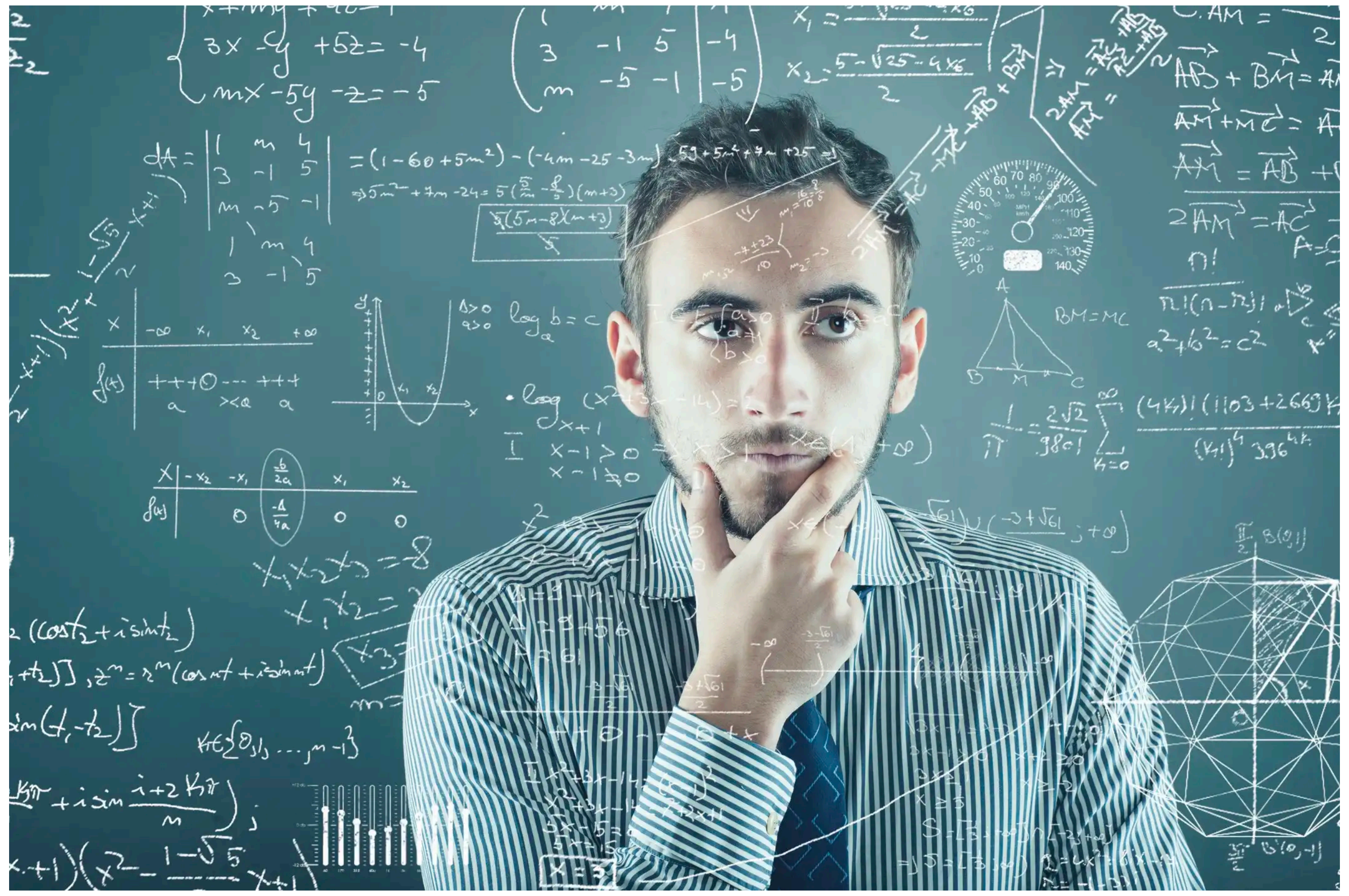
Assembly programmer



C programmer

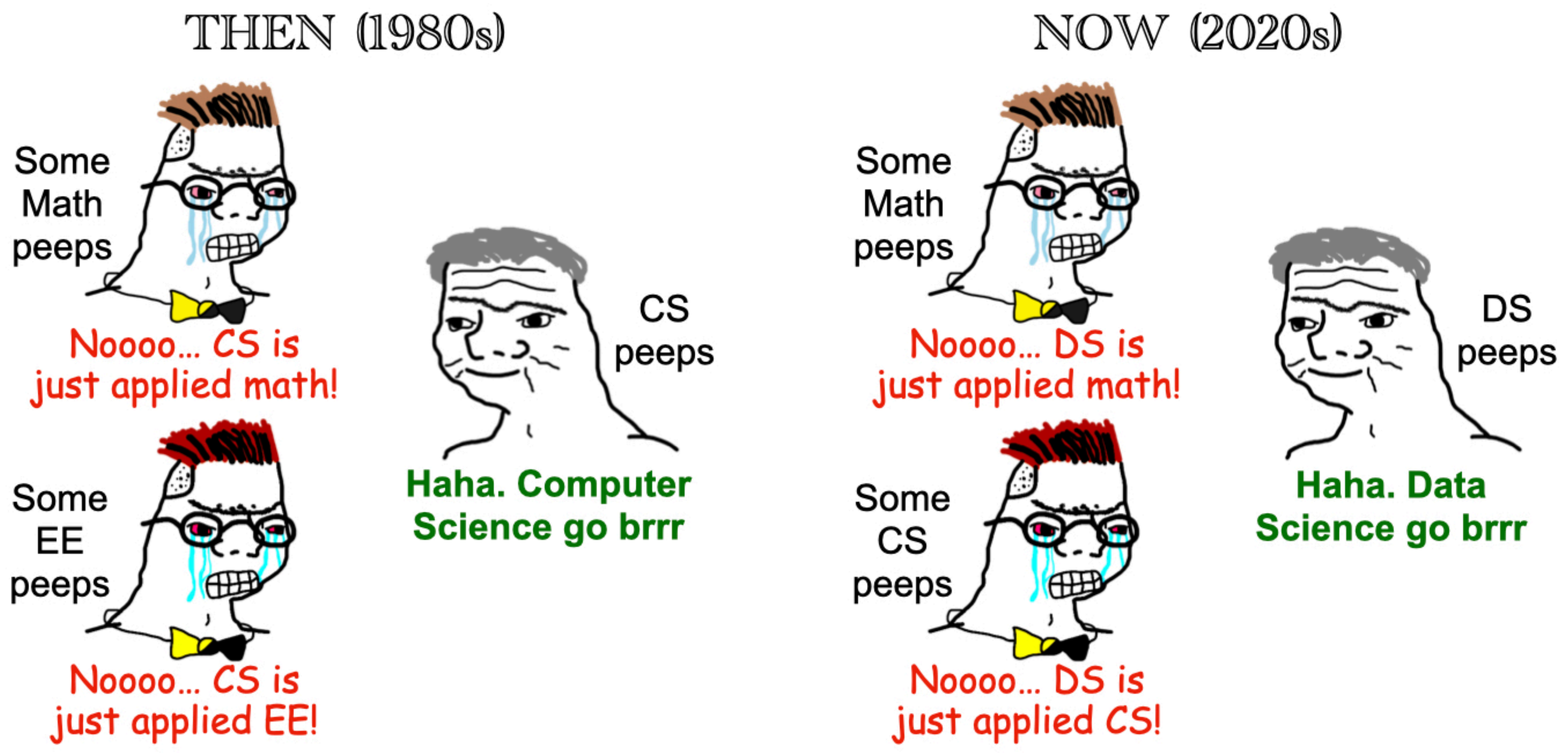
```
#include <stdio.h>
int main(){
int i, n = 10, t1 = 0, t2 = 1, nxt;
for (i = 1; i <= n; ++i){
printf("%d, ", t1);
nxt = t1 + t2;
t1 = t2;
t2 = nxt; }
return 0; }
```

Data science



What is this course about?

Data science professionals ought to be familiarized with data systems from a **user**'s standpoint, as opposed to the conventional approach of a **system implementer**.



What is this course about?

- Relational databases
- NoSQL datastores
- Stream or batch processors
- Message brokers
- Spark, MapReduce, Hadoop, Kafka, HDFS
- Data lakes, column database
- ....

How to use and operate them more effectively?

What is this course about?

- Foundations of data systems
- Scaling distributed systems
- Data Processing and Programming model.
- AI as the New Abstraction for Data Systems

4. AI through Prompts

3. Programming interface

2. Distributed Systems

1. Data systems

Classes in AI Era

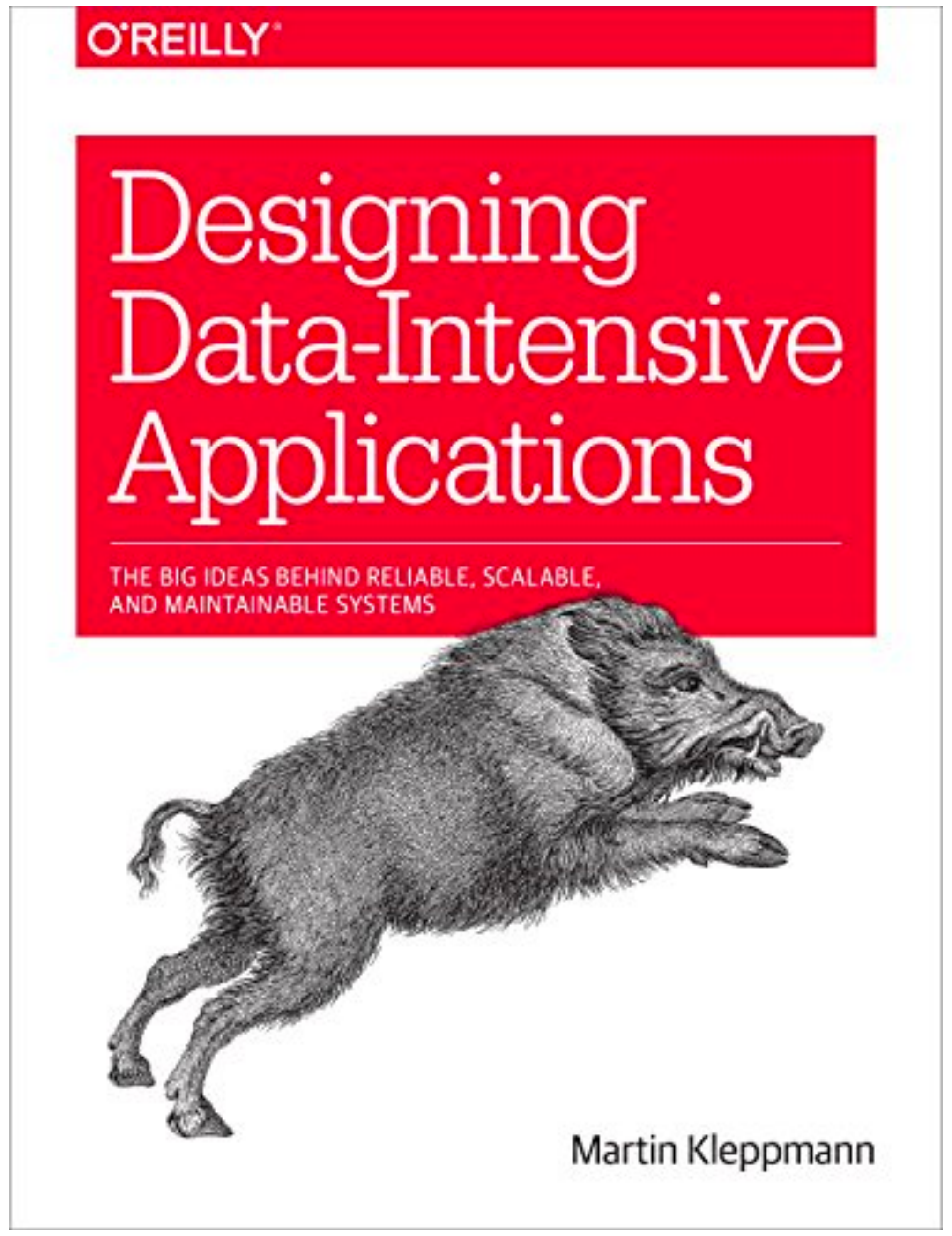
What is this course about?

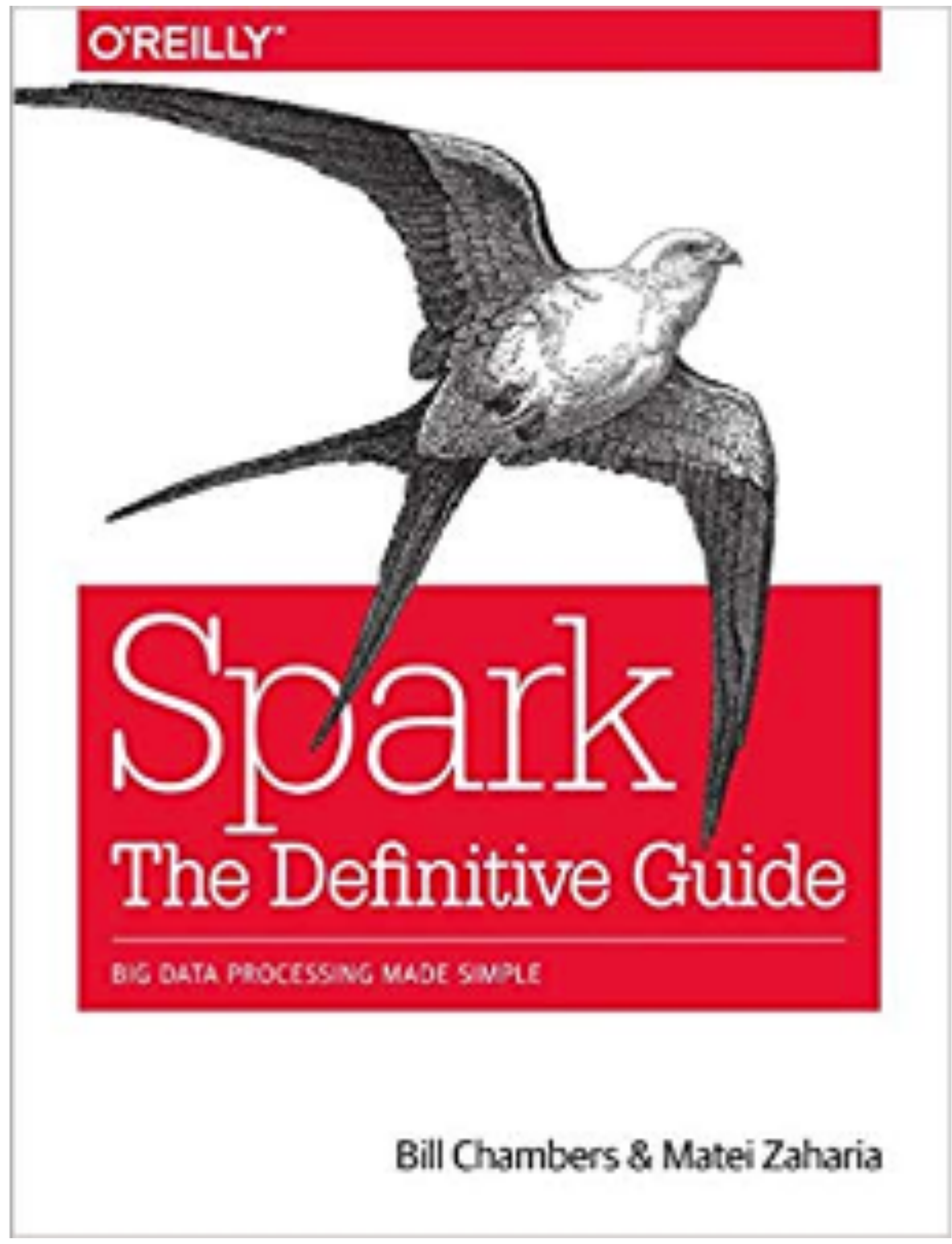
The data sci relevant components in the following course

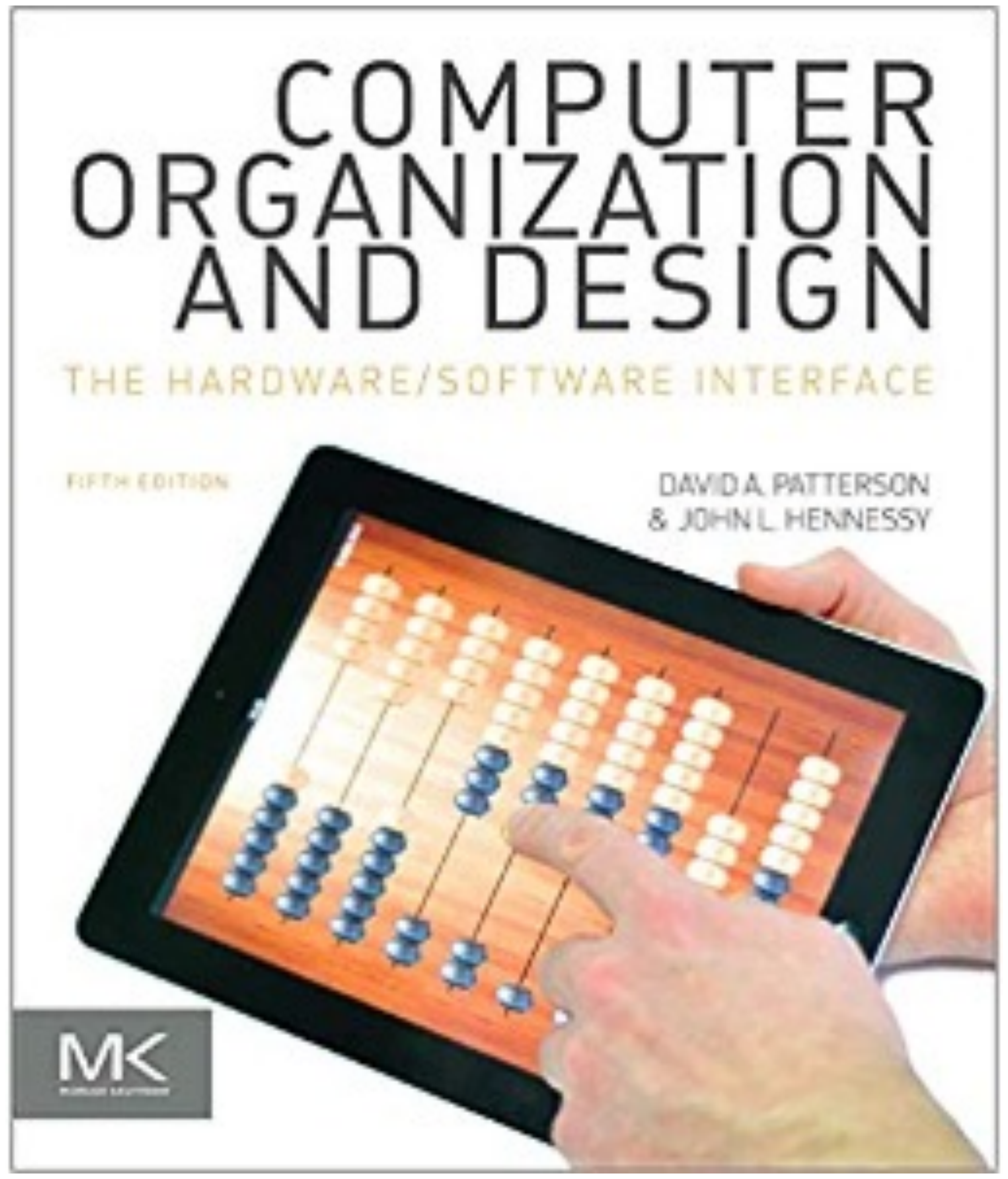
- Computer organization
- System programming
- Networks
- Operating systems
- Distributed systems
- Cloud computing
- + various data sci/AI tricks

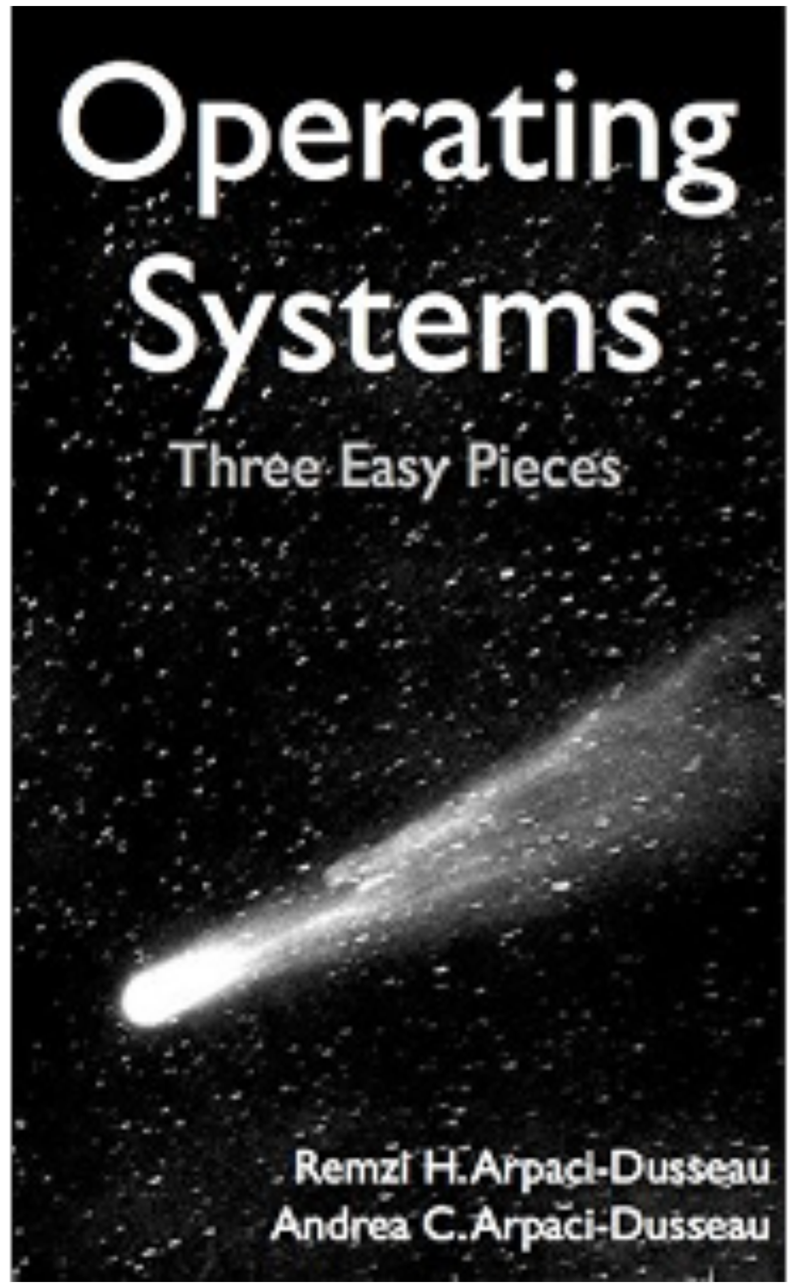
Suggested Textbooks

Computer systems are about carefully layering levels of abstraction.





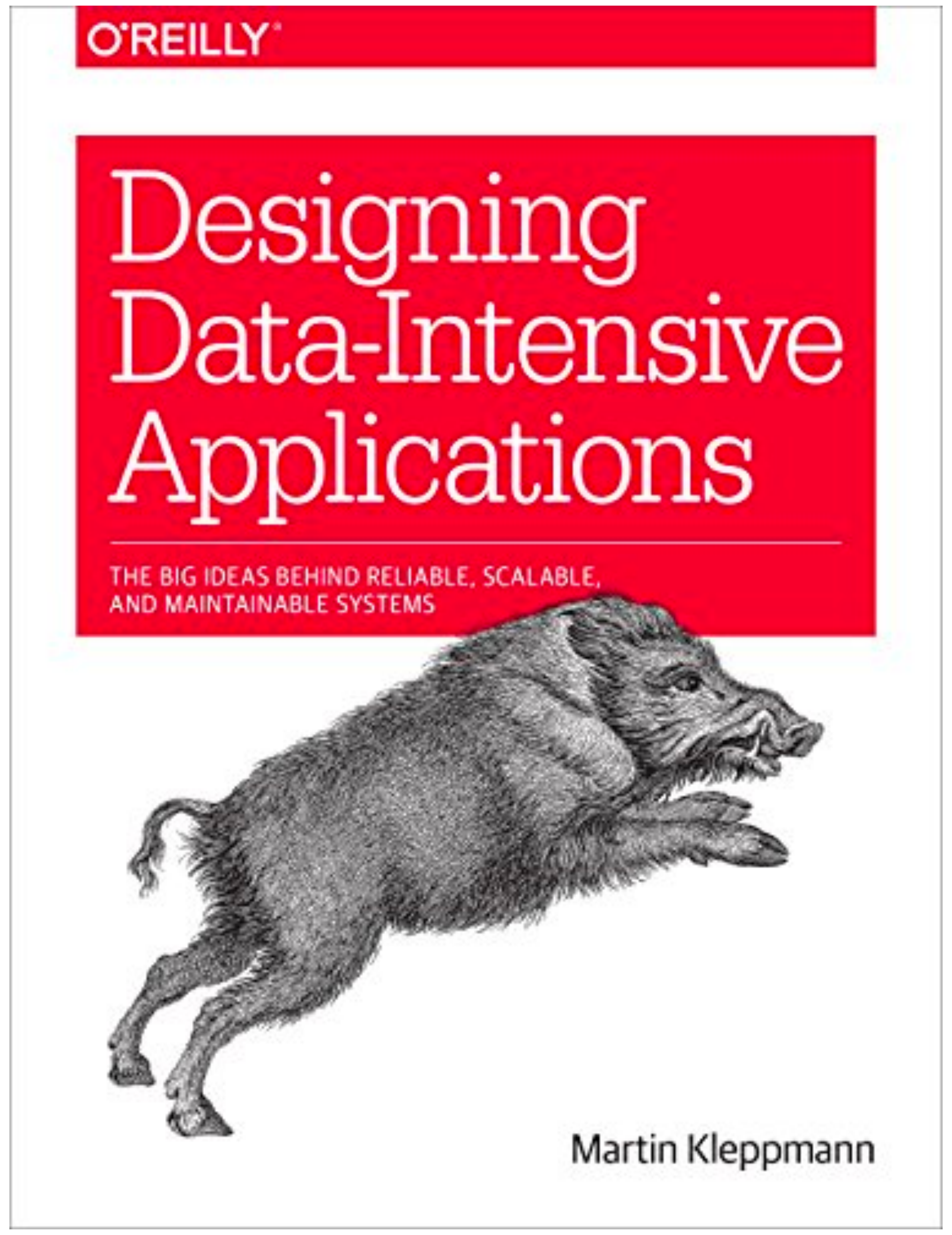




Scalable data flows

Low-level system software

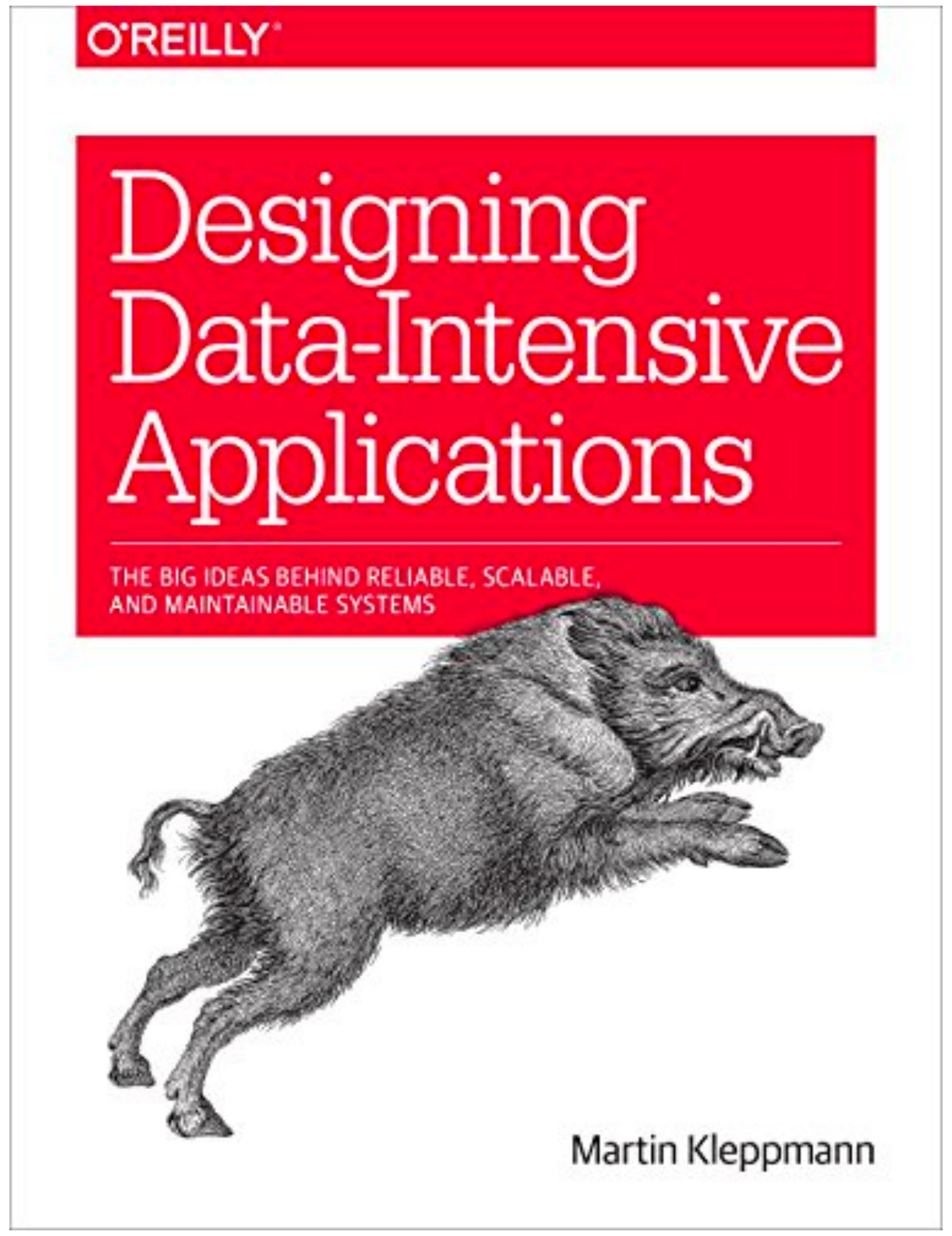
Suggested Textbooks

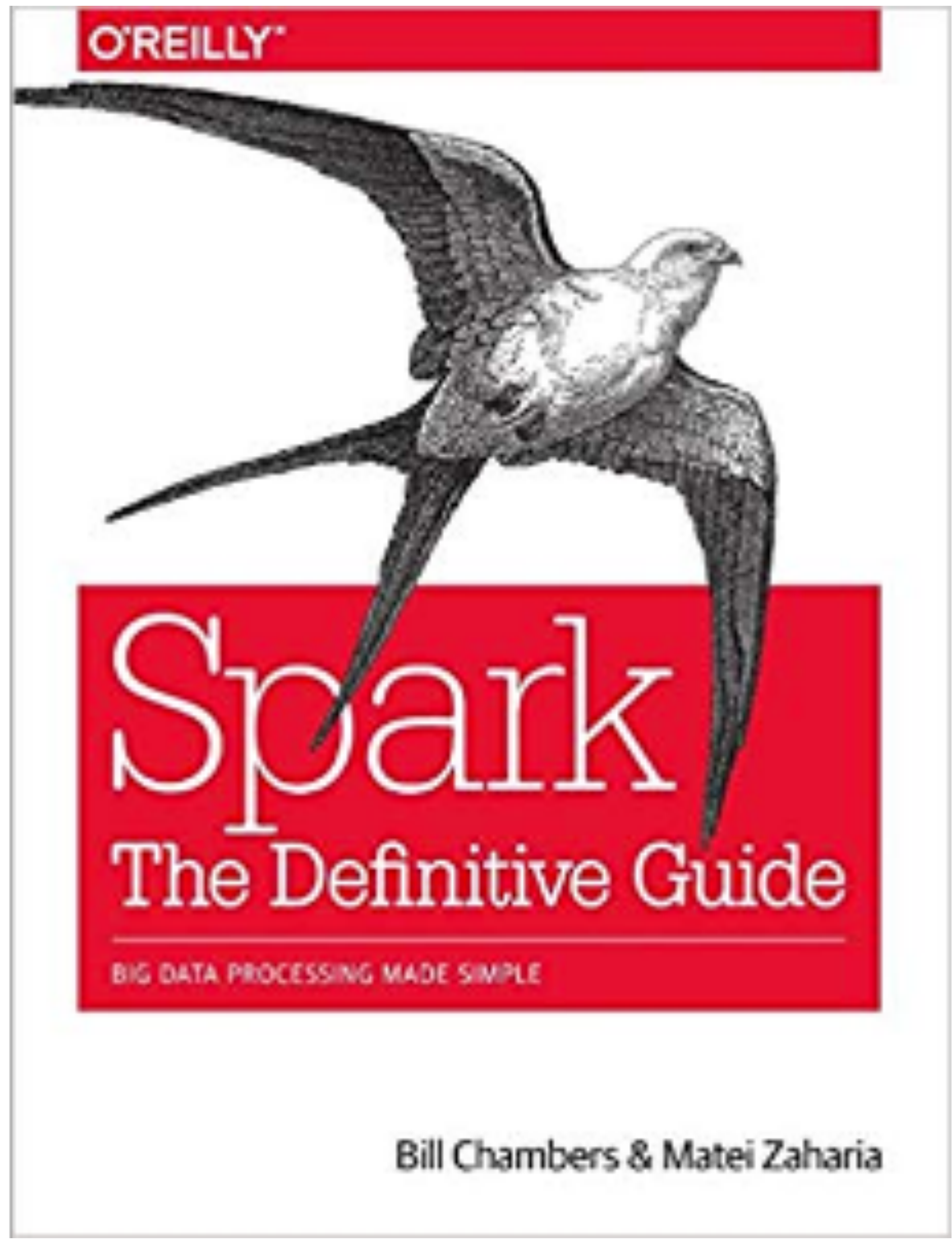


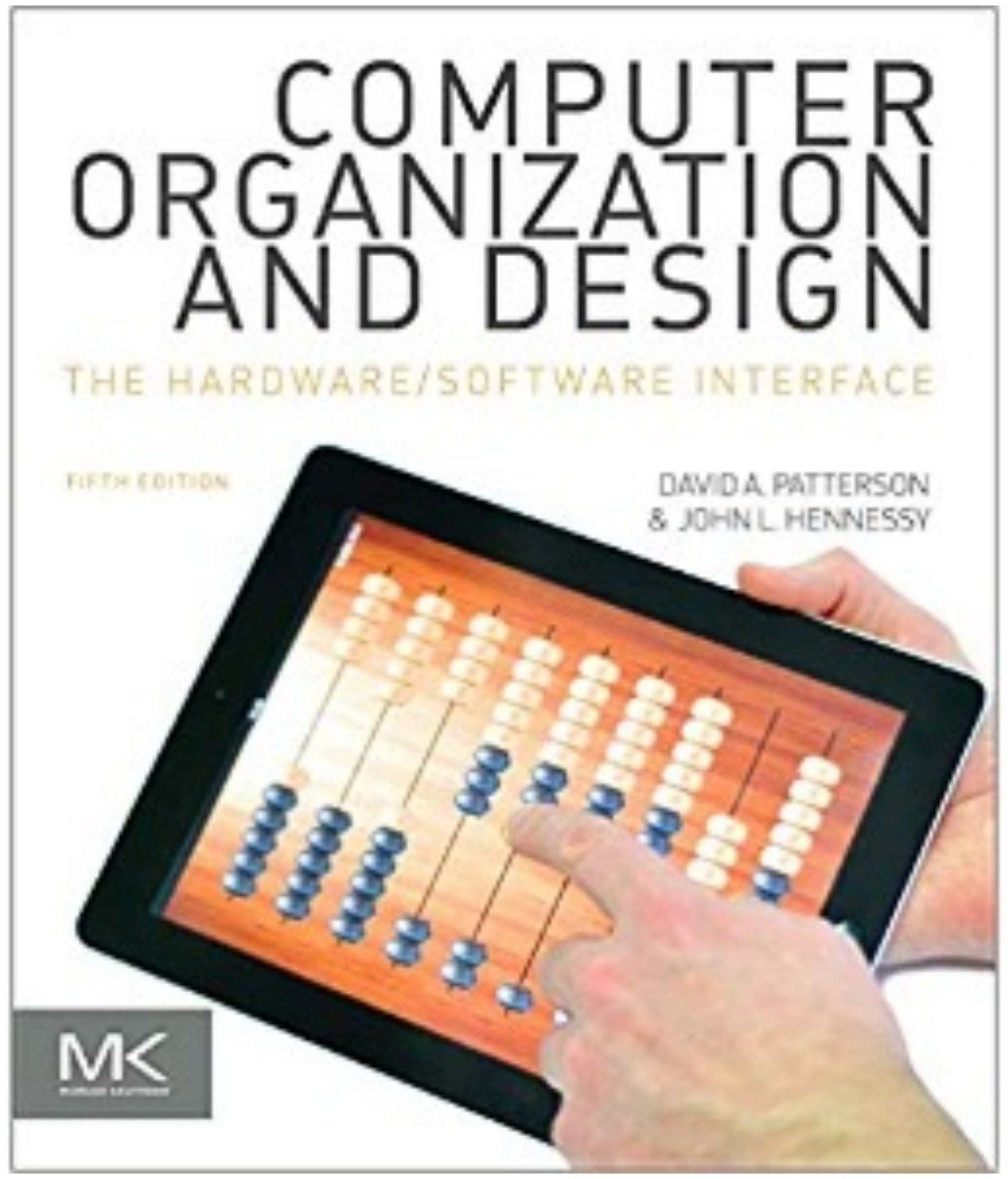
- Chapter 3. Storage and retrieval
- Chapter 4. Encoding and evolution
- Chapter 10. Batch processing
- Chapter 11. Stream processing
- Chapter 12. The future of data systems
- ~~The other chapters~~

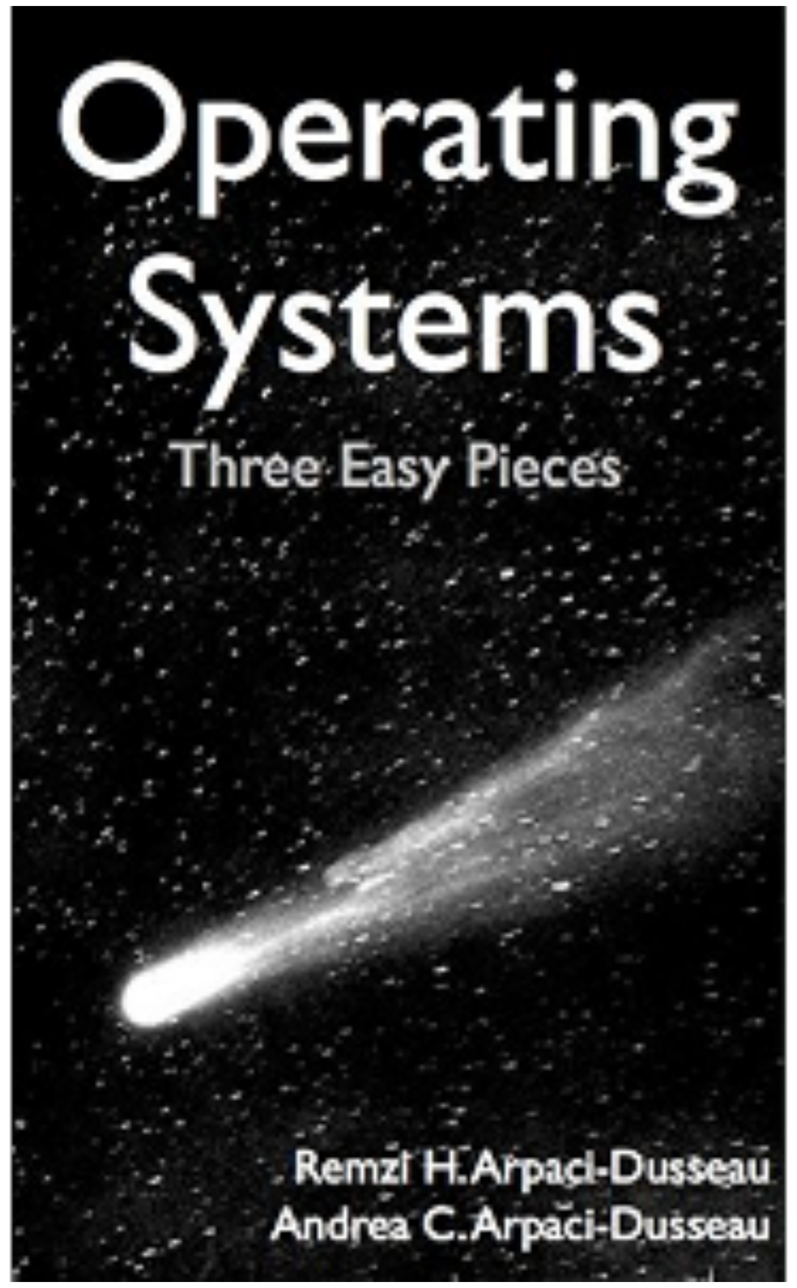
Suggested Textbooks

Computer systems are about carefully layering levels of abstraction.









Hands on experience

Background

AI teachers

The Hero with a Thousand Faces

Shadowing How Peter Steinberger Codes with AI

AI coding is a tool rather than a lootbox.

FEBRUARY 23, 2026 - 5 MINUTE READ - RESEARCH EN

> *I spent 3 hours watching OpenClaw's creator code, frame by frame. Here's what I learned about the gap between amateur and elite AI-assisted programming.*

My first job was at Yahoo. On day one, my manager didn't give me a project. Instead, he told me to shadow a senior engineer with 20+ years of experience. I sat in the cubicle next to the senior engineer for a few hours and watched everything — every keystroke, every mouse movement, every shortcut. When I didn't understand something, I asked: Why this shortcut? Why not use the mouse here? That experience influenced my working habits for many years.

What is this course about?

- Foundations of data systems
  - Data models, big data storage and retrieval, and how to encode information when you store data, etc.
  - ~~Transactions, synchronization, consistency, consensus~~

What is this course about?

- Scaling distributed systems
  - Cluster, cloud, edge, network, replication, partition, consistency, ACID, etc.
  - ~~RPC, Caching, Fault tolerance, Paxos, Concurrency~~

What is this course about?

- **Data Processing and Programming model.**
  - Batch processing, stream processing, MapReduce, Hadoop, Spark, Kafka, etc.

Learning outcomes of this corse

- **Explain** the basic principles of data systems, distributed systems, and data programming model.
- **Identify** the abstract data access patterns of, and opportunities for parallelism and efficiency gains in data processing at scale.
- **Gain** hands-on experience in creating end-to-end pipelines for data preparation, feature engineering, and model selection on large-scale datasets.
- **Reason** critically about practical tradeoffs between accuracy, runtimes, scalability, usability, and total cost.

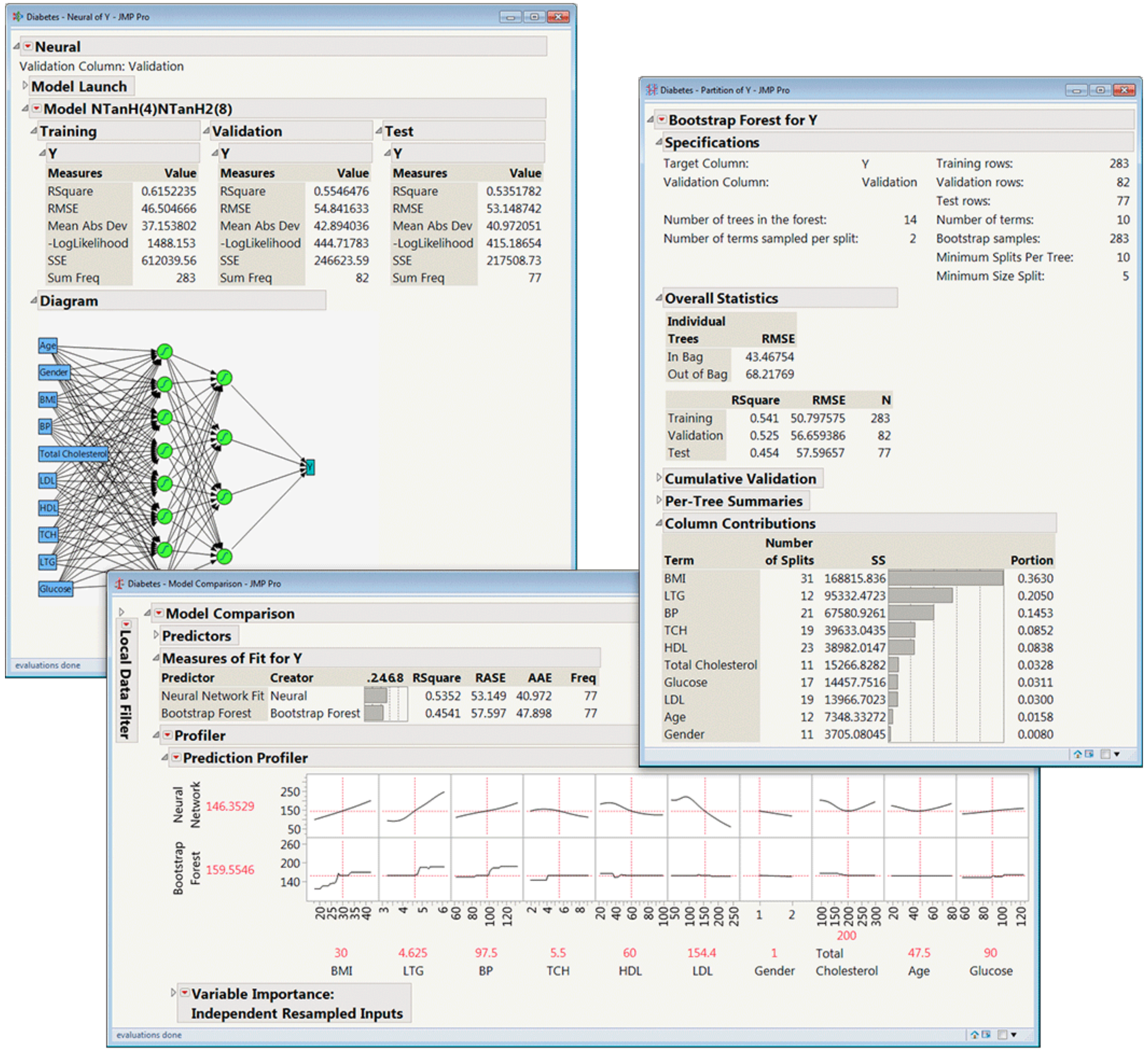
What this course is **NOT** about

- Not a course on database, relational model, or SQL
  - Take DSC 202 instead (pre-requisite)
- Not a course on how to build scalable data systems
  - Take Distributed Systems, Operating Systems, Cloud Computing, …
- Not a training module for how to use Spark
  - We focus more on principles.
- If you have taken DSC 102 and look for a graduate version
  - Take DSC 204A next Winter.

Why bother learning such low-level computer sciencey stuff in Data Science?

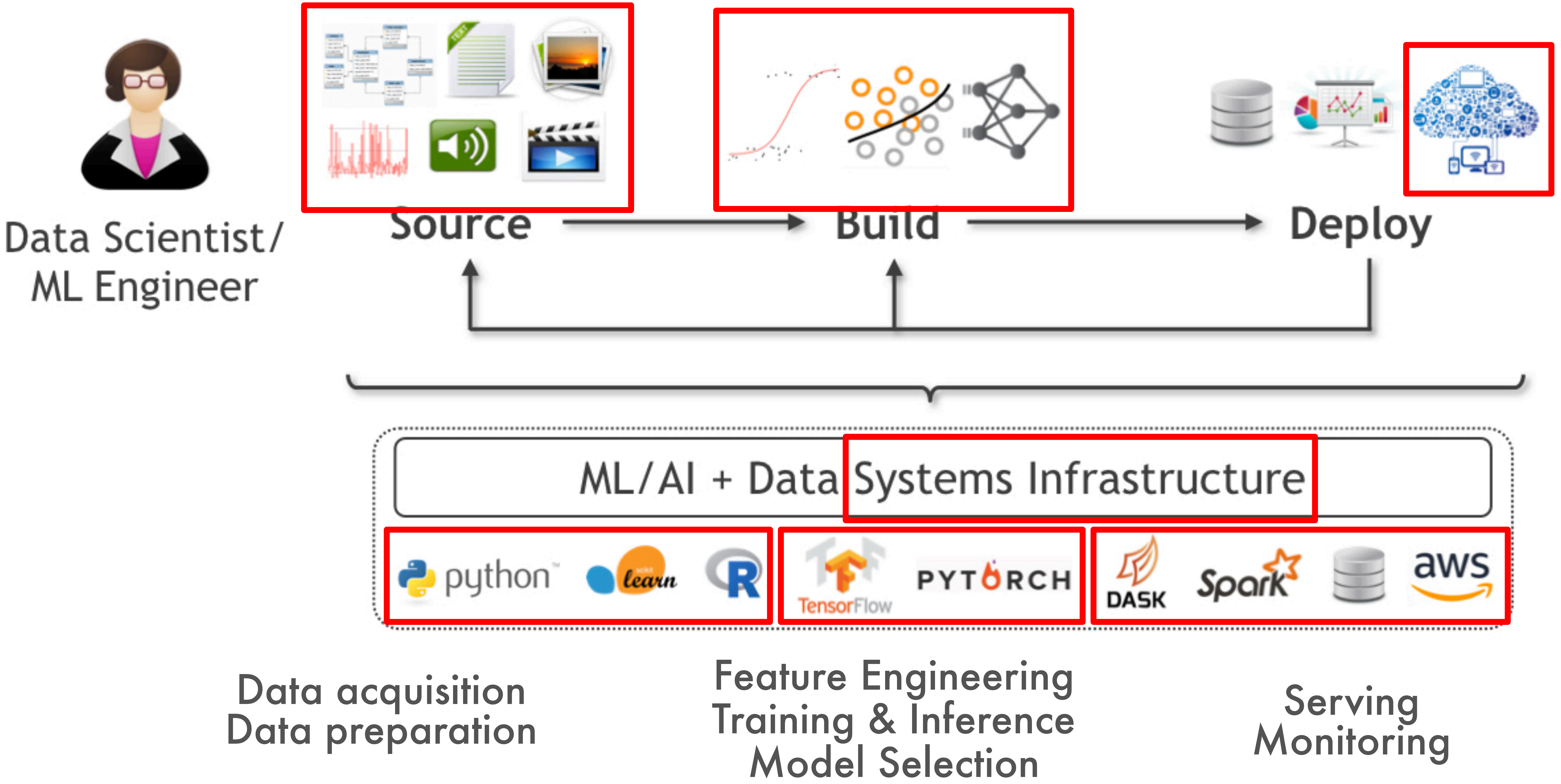
Luxury of "Statisticians"/"Analysts"

- Methods: Sufficed to learn just math/stats, maybe some SQL
- Types: Mostly tabular (relational), maybe some time series
- Scale: Mostly small (KBs to few GBs)
- Tools: Simple GUIs for both analysis and deployment; maybe an R-like console



https://www.jmp.com/en_au/offers/jmp-pro-for-academic-research.html
https://www.technologymagazine.com/data-and-data-analytics/sas-tops-worldwide-advanced-and-predictive-analytics-market-share

Reality of Today's "Data Scientists"



glassdoor

statistician | Location

Statistician Salaries

United States

Overview | **Salaries** | Interviews | Insights | Career Path

How much does a Statistician make?

Updated Jan 4, 2022

Industry: All industries

Employer Size: All company sizes

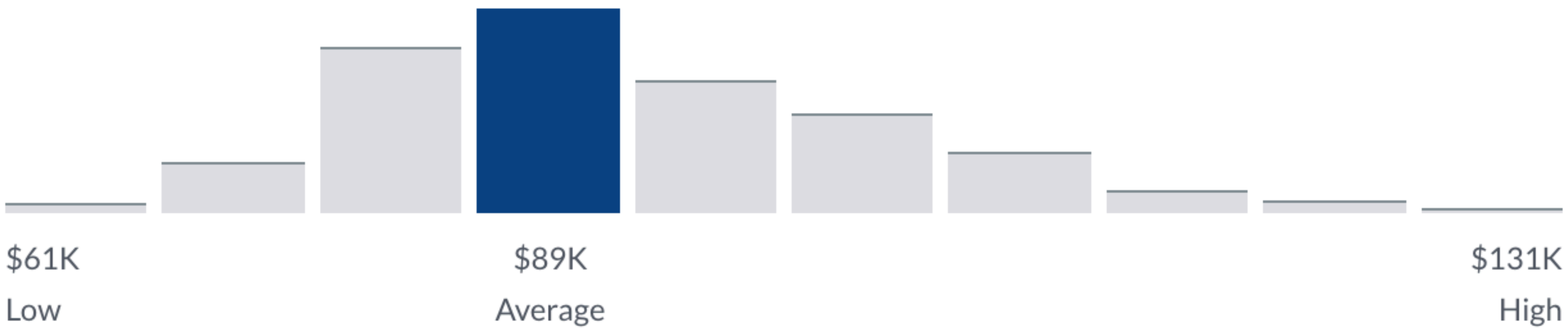
Experience: All years of Experience

Very High Confidence

$88,989 /yr

Average Base Pay

2,398 salaries



glassdoor

data scientist | Location

Data Scientist Salaries United States

Overview | **Salaries** | Interviews | Insights | Career Path

How much does a Data Scientist make?

Updated Jan 4, 2022

Industry: All industries

Employer Size: All company sizes

Experience: All years of Experience

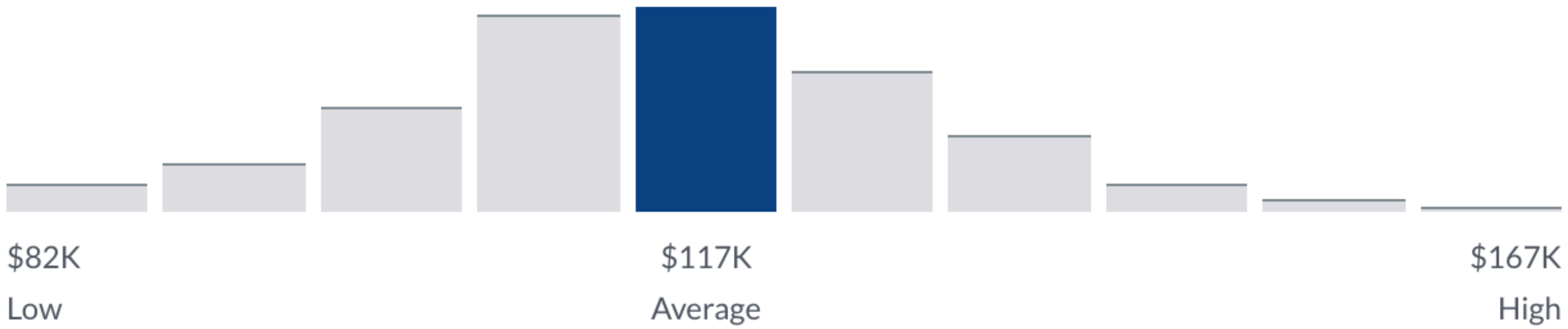
To filter salaries for Data Scientist, **Sign In** or **Register**.

Very High Confidence

$117,212 /yr

Average Base Pay

18,354 salaries



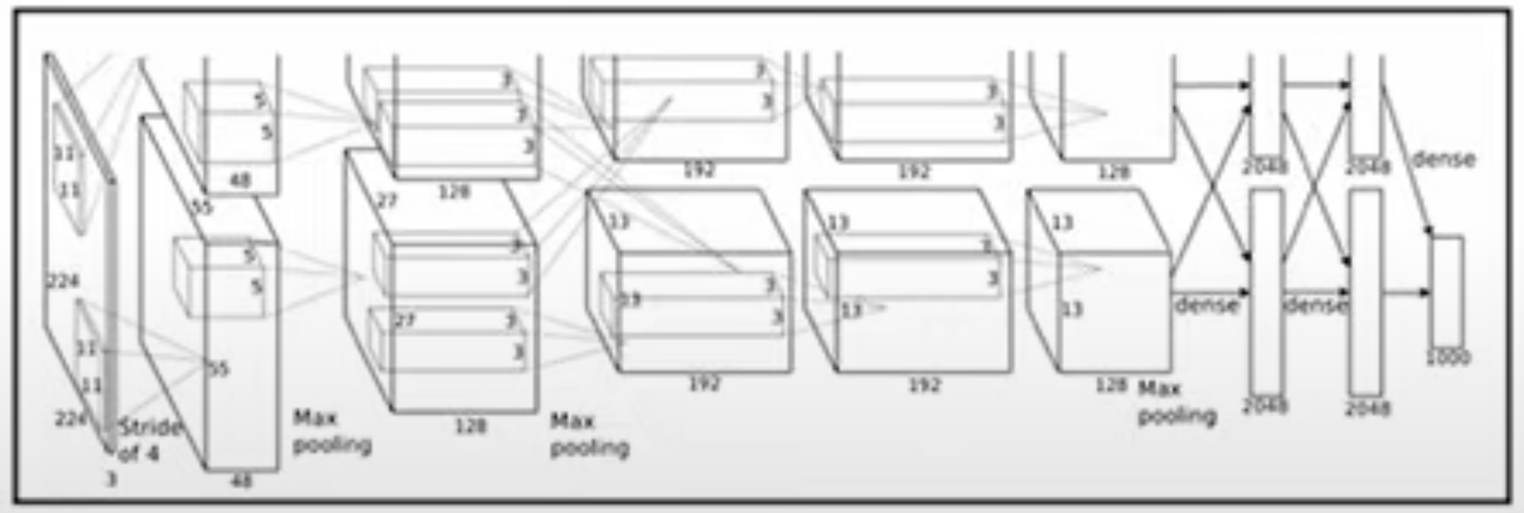
— 88,989

= 28,223!

AI as the new abstraction

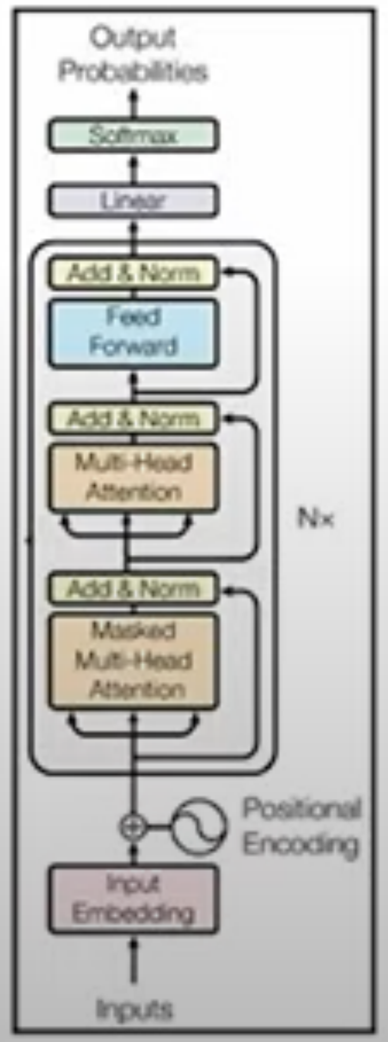
Software 1.0	Software 2.0	Software 3.0
computer code	weights	prompts
↓ programs	↓ programs	↓ programs
computer	neural net	LLM

became programmable in **~1940s**



fixed function neural net

e.g. AlexNet: for image recognition (**~2012**)



Questions?

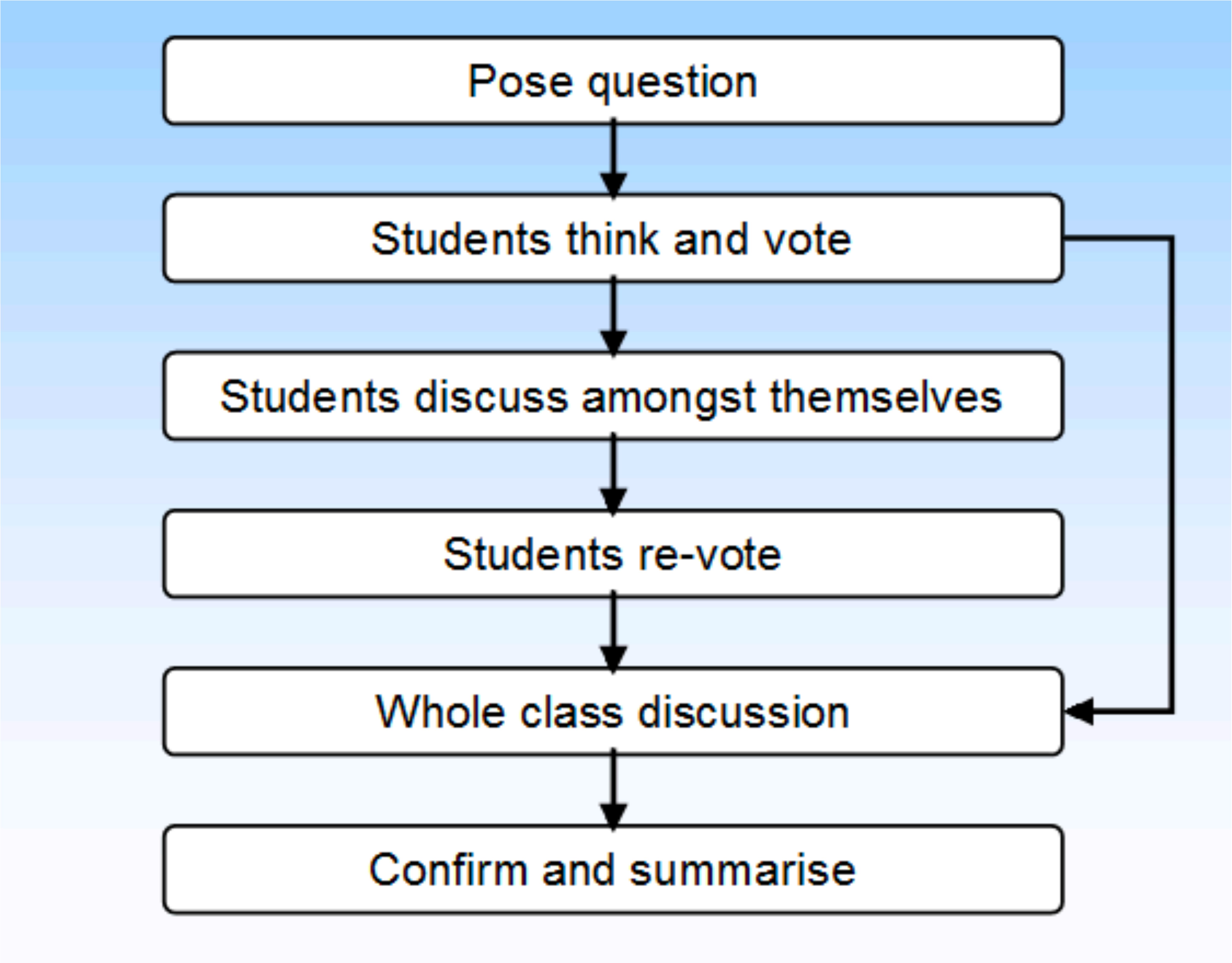
Prerequisites

- DSC 200, 202 (or equivalent).
- Proficiency in Python programming & Terminals
- Network basics
- For all other cases, email me with proper justification; a waiver can be considered

Components and Grading

- 5 Programming Assignments: **40%** (8%x5)
  - Encourage you to use AI tools
  - No late days! Plan your work well ahead.
- Final Exam (TBD): **40% ?**
- Peer Instruction Activities: **20%**
  - **Students can opt out, in which case the Final Exam weight becomes 60%.**
- Extra Credit Peer Evaluation Activities: **4%** (likely)

Peer instruction activity



Grading Scheme (grade is the better of the two)

Grade	Absolute Cutoff (>=)	Relative Bin (Use strictest)
A+	95	Highest 5%
A	90	Next 10% (5-15)
A-	85	Next 15% (15-30)
B+	80	Next 15% (30-45)
B	75	Next 15% (45-60)
B-	70	Next 15% (60-75)
C+	65	Next 5% (75-80)
C	60	Next 5% (80-85)
C-	55	Next 5% (85-90)
D	50	Next 5% (90-95)
F	< 50	Lowest 5%

Grading Scheme (grade is the better of the two)

Grade	Absolute Cutoff (>=)	Relative Bin (Use strictest)
A+	95	Highest 5%
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B	75	Next 15% (45-60)
B-	70	Next 15% (60-75)
C+	65	Next 5% (75-80)
C	60	Next 5% (80-85)
C-	55	Next 5% (85-90)
D	50	Next 5% (90-95)
F	< 50	Lowest 5%

Example, 82 and 33%,

Rel: B-; Abs: B+;

Final: B+

The structure of the course

	Topics	
Week 1-3	Foundations of Data Systems	Single Machine: CompOrg -> OS -> Cloud
Week 4-6	Scaling Distributed Systems	Multiple Machine: Storage -> Network
Week 7-10	Data Processing and Programming model	Processing: Batch -> Stream -> Cloud

https://haojian.github.io/DSC204A26SP/

Programming Assignments

- PA0: Setting up AWS and Dask
  - Apr 10 to Apr 25
- PA1: Data Exploration with Dask
  - Apr 26 to May 10
- PA2: Feature Eng. and Model Selection with Spark
  - May 11 to June 2
- You only have $50 AWS credit! Close the instance when you finish.

Expectations on the PAs

- Expectations on the PAs:
  - Individual projects; see webpage on academic integrity
- I will cover the concepts and tools' tradeoffs in the lectures
- TAs will explain and demo the tools; handle all Q&A
- You are expected to put in the effort to learn the details of the tools' APIs using their documentation on your own!

Respecting TAs’ time

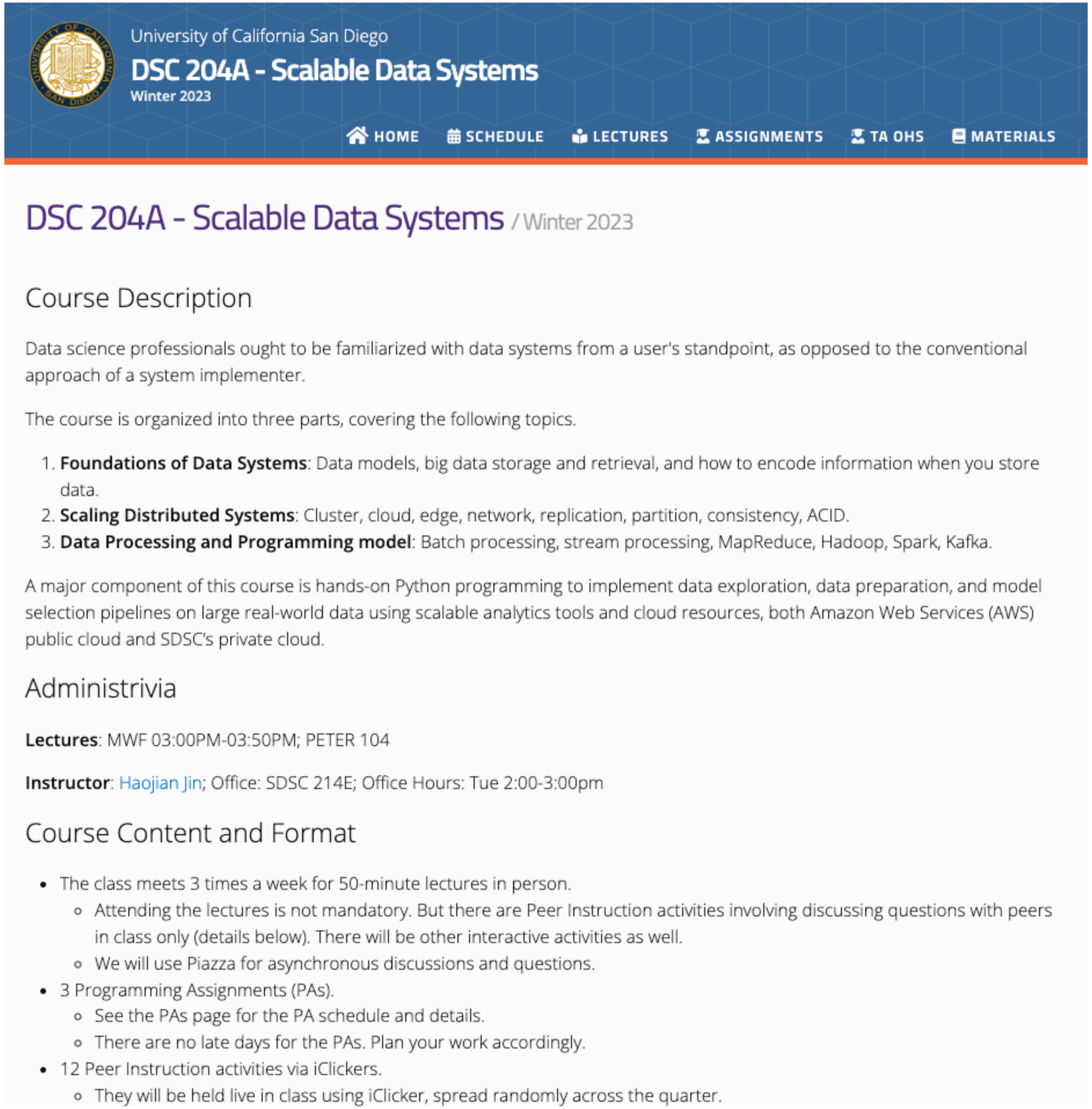
- Office hours are for getting ideas on how to debug or better approach your homework.
- Write a description! Try to narrow down your problem area as much as possible.
- If you don’t have a description, TA can reject your questions.
- Respect TA’s working hours.
  - Respond in 24 hours.
  - Members may send msgs at night or on weekends, but only expect to receive a reply on weekday.

Tentative plan

- Rohit
  - Tuesday 1:30 PM - 2:30 PM.
- Megha
  - Thursday TBD.
- Location:
  - CSE building or HDSI building?

Course administrivia

https://haojian.github.io/DSC204A26SP


University of California San Diego

DSC 204A - Scalable Data Systems

Winter 2023

HOME SCHEDULE LECTURES ASSIGNMENTS TA OHS MATERIALS

DSC 204A - Scalable Data Systems / Winter 2023

Course Description

Data science professionals ought to be familiarized with data systems from a user's standpoint, as opposed to the conventional approach of a system implementer.

The course is organized into three parts, covering the following topics.

1. **Foundations of Data Systems**: Data models, big data storage and retrieval, and how to encode information when you store data.
2. **Scaling Distributed Systems**: Cluster, cloud, edge, network, replication, partition, consistency, ACID.
3. **Data Processing and Programming model**: Batch processing, stream processing, MapReduce, Hadoop, Spark, Kafka.

A major component of this course is hands-on Python programming to implement data exploration, data preparation, and model selection pipelines on large real-world data using scalable analytics tools and cloud resources, both Amazon Web Services (AWS) public cloud and SDSC's private cloud.

Administrivia

Lectures: MWF 03:00PM-03:50PM; PETER 104

Instructor: Haojian Jin; Office: SDSC 214E; Office Hours: Tue 2:00-3:00pm

Course Content and Format

- The class meets 3 times a week for 50-minute lectures in person.
  - Attending the lectures is not mandatory. But there are Peer Instruction activities involving discussing questions with peers in class only (details below). There will be other interactive activities as well.
  - We will use Piazza for asynchronous discussions and questions.
- 3 Programming Assignments (PAs).
  - See the PAs page for the PA schedule and details.
  - There are no late days for the PAs. Plan your work accordingly.
- 12 Peer Instruction activities via iClickers.
  - They will be held live in class using iClicker, spread randomly across the quarter.

Course administrivia

- Lectures: MWF 3pm-3:50pm PT at PETER 104
- Instructor: Haojian Jin; haojian@ucsd.edu
  - OHs: TBD. See website.
- TAs: Rohit Ramaprasad; Megha Agarwal
  - TA hours see the course web site.
- Slack for all communications (also see Canvas).
- Canvas for PA submission, Peer Evaluation Activities, Grading.

General Dos and Do NOTs

- Do:
  - Follow all announcements on Piazza
  - Try to join the lectures/discussions live
  - Participate in discussions in class / on Piazza
  - Raise your hand before speaking
  - View/review podcast videos asynchronously by yourself
  - To contact me/TAs, use Slack first; if you really need to email, use "DSC 204:" as subject prefix

General Dos and Do NOTs

- Do NOT:
  - Harass, intimidate, or intentionally talk over others
  - **Violate academic integrity** on the PAs, exams, or other components; I am very strict on this matter!

Questions?